Assignment 2:

Write a C program that includes a user-defined function named isArmstrong with the signature int isArmstrong(int num);. An Armstrong number is a number that is equal to the sum of its own digits each raised to the power of the number of digits. For example, 153 is an Armstrong number because 1^3 + 5^3 + 3^3 = 153

Source Code:

```
#include<stdio.h>
#include<math.h>

int isArmstrong(int num) {
   int digits=0,sum=0,r;
   for (int m=num;m>0;m/=10) {
      digits++;
   }
   for (int m=num;m>0;m/=10) {
      r=m%10;
      sum=sum+pow(r,digits);
   }
   return sum==num;
}

int main() {
   int n;
   printf("Enter the number : ");
   scanf("%d",&n);
   if (isArmstrong(n))
      printf("This is an Armstrong number.");
   else
      printf("This is not an Armstrong number.");
   return 0;
}
```

Output:

```
Enter the number :153

This is an Armstrong number.
Process finished with exit code 0
```

```
Enter the number :762

This is not an Armstrong number.
Process finished with exit code 0
```